NATIONAL UNIVERSITY OF SINGAPORE
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Experiment-Based Risk Assessment Form

Name of Department	Department of Chemistry	**Name and Location of Lab**	S5-01
Name of Laboratory	General and Synthesis Chemistry Lab	**Name of PI (lecturer-in-charge)**	Dr. Edith S. H. Chan
Name of Student	Kim Donghwan	**Name of Activity/Experiment**	CM3191 Inorganic - Metallocene: Session II Purification of Ferrocene by Sublimation and Characterization

Hazard Identification				Risk Evaluation & Control						
No	**Description / Details of Steps in Activity**	**Hazard(s)**	**Possible Accident(s)**	**Existing Risk Control (Mitigation)**	**Severity**	**Likelihood (probability)**	**Risk Level**	**Additional Risk Control**	**Person Responsible**	**By (Date)**
1	Weigh the two Petri dish covers and the crude ferrocene from Session I. Spread the crude solid thinly on the small Petri dish base.	Chemical hazard: Ferrocene is a flammable solid, harmful if swallowed or inhaled, may damage fertility or the unborn child, and may cause liver damage through prolonged or repeated inhalation.	Inhalation of dust. Harmful if swallowed. Contamination of hands and balance area.	1. Handle the solid in a well ventilated fumehood; do not breathe dust. 2. Wear suitable PPE (lab coat, goggles, gloves and covered shoes). 3. Transfer with a spatula over a tray; keep the container closed. 4. Wash hands after handling. No food or drink in the lab.	2	1	2			
		Broken glassware (Petri dishes, vials).	Cuts from broken glassware.	Visual inspection for any cracks before usage. Proper disposal of broken glassware in sharps container.	2	1	2			
2	Place the Petri dish assembly on the hotplate in the fumehood and heat slowly to below 150 °C (100 °C for the first cycle).	Heat hazard: Hotplate and Petri dish are hot (up to 150 °C).	Burns from hot surfaces or hot glassware. Glass may crack on heating.	1. Conduct the sublimation in a well ventilated fumehood. 2. Raise the temperature slowly; do not exceed 150 °C. 3. Use heat resistant gloves or tongs for the hot cover; place it on a heat resistant mat. 4. Do not leave the hotplate unattended.	2	2	4	Run the first cycle at 100 °C (crude is still wet). Check the set point before each cycle.	Dr. Chan	17 Sep 2026
		Chemical and fire hazard: Ferrocene is a flammable solid and is heated close to its melting point. Fine dust may ignite.	Ignition of ferrocene vapour or dust. Sample melts if overheated. Combustion gives carbon oxides and iron oxides.	1. Keep away from open flames and other ignition sources. 2. Do not exceed 150 °C. 3. Keep the crude and the collected product away from the hotplate. 4. In case of fire, use dry sand, dry chemical or alcohol resistant foam to extinguish.	2	1	2			
3	Place a 400 mL beaker of ice water on the Petri dish cover as the cold surface. Remove it between cycles.	Thermal shock; water next to the hotplate.	Cover cracks. Water spilled onto the hotplate causes steam burns or an electrical fault.	1. Inspect the cover for cracks before use. 2. Fill the beaker about half full; lower it gently, do not slide it. 3. Wipe condensation off the beaker before lifting it. 4. Keep the hotplate cable away from water. 5. Avoid reaching over the hotplate when handling the beaker.	2	1	2			

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4	Cool the cover, scrape off the sublimed ferrocene, weigh, and repeat the cycle until no more solid sublimes.	Heat and chemical hazard: Hot glass; ferrocene dust raised by scraping.	Burns from a cover that is still hot. Inhalation of ferrocene vapour or dust. Cuts if the spatula slips.	1. Lift the cover inside the fumehood. 2. Allow the cover to cool to room temperature before handling; do not cool it with water. 3. Scrape gently in the fumehood over the collection vessel, away from the hotplate. 4. Wear suitable PPE (lab coat, goggles, gloves and covered shoes).	2	1	**2**			
5	Examine the crude and purified ferrocene under the stereomicroscope (14x).	Electrical hazard; warm lamp housing.	Electric shock from a damaged cable. Minor burn.	1. Check the cable and plug before use. 2. Present the sample on a watch glass; wipe the stage afterwards.	1	1	**1**			
6	Grind 0.09-0.12 g of ferrocene in an agate mortar. Pack into the MSB tube (15-25 mm) and measure on the magnetic susceptibility balance.	Chemical hazard: Grinding produces fine ferrocene dust (see step 1).	Inhalation of dust.	1. Grind in the fumehood with the sash low; grind gently, away from the hotplate. 2. Wear suitable PPE (lab coat, goggles, gloves and covered shoes). 3. Clean the mortar and bench with a damp tissue; dispose in solid waste.	2	1	**2**			
		Broken glassware; magnetic field of the balance.	MSB tube snaps during packing, causing cuts. Strong magnet affects cards and metal objects.	1. Pack by tapping the tube on the bench; do not force with a rod. 2. Keep metal objects and cards away from the balance.	2	1	**2**			
7	Dissolve 35-40 mg of ferrocene in methanol in a 25 mL volumetric flask. Dilute in the fumehood and record the UV-Vis spectrum (300-700 nm).	Chemical and fire hazards: Methanol is highly flammable (flash point 9.7 °C), toxic if swallowed, inhaled or in contact with skin, and causes damage to the eyes and central nervous system.	Splash to eyes or skin. Inhalation of vapour. Ignition of vapour at the hotplate.	1. Handle methanol in a well ventilated fumehood. 2. Wear suitable PPE (lab coat, goggles, gloves and covered shoes). Use a pipette filler. Change gloves if they get wet. 3. Keep methanol away from the hotplate and other ignition sources; keep the bottle closed. 4. In case of fire, use dry sand, dry chemical or alcohol resistant foam to extinguish.	3	1	**3**	Switch off the hotplate, or use a separate fumehood, before opening methanol.	**Dr. Chan**	**17 Sep 2026**
8	Record the solid IR spectrum (ATR or KBr disc).	Chemical hazard: KBr causes serious eye irritation. Mechanical and electrical hazards: press / ATR anvil; instrument.	KBr dust in eyes. Pinch injury from the press or anvil. Electric shock from a damaged cable.	1. Wear goggles. 2. Keep fingers clear of the press platens and anvil; tighten the anvil only to the indicated pressure. 3. Check the cable and plug before use.	2	1	**2**			

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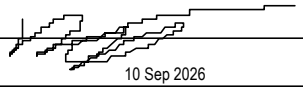
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9	Dissolve about 1 mg of ferrocene in 0.6 mL CDCl3 in an NMR tube. Cap and label the tube.	Chemical hazard: Deuterochloroform is toxic if inhaled, harmful if swallowed, irritating to skin and eyes, suspected of causing cancer and of damaging fertility, and causes liver and kidney damage on repeated exposure.	Inhalation of vapour. Skin contact. Cuts from a broken NMR tube.	1. Dispense CDCl3 only in the fumehood; cap the tube immediately. 2. Wear suitable PPE (lab coat, goggles, gloves and covered shoes). Change gloves if they get wet. 3. Use a glass Pasteur pipette. 4. Chlorinated waste to the chlorinated solvent container.	2	1	**2**			
10	Weigh 1-2 mg of ferrocene into a sample tube for ESI-MS (solvent added at the instrument).	Chemical hazard: Ferrocene (see step 1). Acetonitrile is highly flammable, harmful and an eye irritant.	Inhalation of dust. Eye irritation from solvent splash.	1. Weigh over a tray with the draught shield closed. 2. Label the tube before adding the sample. 3. If solvent is added by the student, do so in the fumehood.	1	1	**1**			

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11	Waste disposal and clean-up.	Chemical and environmental hazard: Ferrocene is very toxic to aquatic life. Mixed solvent waste.	Release of ferrocene to the drain. Mixing chlorinated and non-chlorinated waste. Cuts from broken glass.	1. Solid ferrocene and contaminated tissues to solid waste; nothing down the sink. 2. Chlorinated and non-chlorinated solvent waste to separate labelled containers; aqueous metal waste to the designated container. 3. Broken glass to the sharps container. 4. Switch off the hotplate and let it cool before clearing the fumehood.	2	1	**2**			

Conducted By

Name Kim Donghwan

Signature 

date 10 Sep 2026

Approved By

Name

Signature

Approval date

Next Revision date (Maximum 3 years)

Hazard information from the SDSs (Sigma-Aldrich) in Canvas Files - Inorganic Part.